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TOPOLOGICAL PHASES OF NEUTRONS – 19.12.2025

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1 Abstract

In this experiment, the geometric phase of a spin-½ neutron was investigated using neutron polarimetry in a Ramsey-type setup. Phase shifts were extracted from sinusoidal intensity oscillations for different rotation schemes of the spin-flip coils and analyzed as a function of the relative rotation angle. While a linear phase dependence was observed, the extracted slopes deviated from the theoretical value of 2. These deviations are attributed to residual dynamical phase contributions arising from an imperfect alignment of the neutron beam with respect to the rotation axes of the coils, leading to unequal path lengths between the spin rotators.

2 Introduction

2.1 Geometric phases

In quantum mechanics, the phase of a wavefunction is generally not observable. However, phase differences can be detected in interference experiments. During the evolution of a quantum state, the accumulated phase can be separated into two contributions: a dynamical phase, which depends on the energy and the duration of the evolution, and a geometric phase, which arises from the geometry of the evolution itself and is independent of the specific dynamics.

Intuitively, the geometric phase can be understood by representing the spin state of a particle as a vector on the Bloch sphere. When the spin is manipulated by external magnetic fields, this vector moves along the surface of the sphere. If the spin undergoes a cyclic evolution along different rotation axes, the resulting path encloses a finite surface area on the Bloch sphere. The geometric phase is a direct consequence of this enclosed area and depends only on the geometry of the path, not on how fast or by which specific fields the evolution is performed.

The geometric phase was introduced by Berry for adiabatic and cyclic evolutions and is therefore commonly referred to as the Berry phase. For a spin-½ system, the geometric phase is given by minus one half of the solid angle enclosed by the evolution path on the Bloch sphere. Consequently, the geometric phase depends solely on the geometry of the path and is independent of external parameters such as time or magnetic field strength.

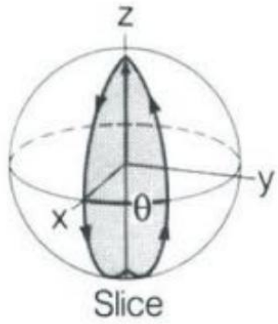

Figure 1 Evolution path of a spin-½ state on the Bloch sphere enclosing a finite solid angle. The geometric phase acquired during the cyclic evolution is proportional to the enclosed area.

2.2 Measurement

The geometric phase cannot be measured as an absolute quantity, but only as a relative phase difference. In this experiment, neutron polarimetry is used to realize a Ramsey-type interferometer for neutron spins. Polarized neutrons are manipulated by a sequence of DC magnetic coils acting as π/2- and π-spin-rotators. The initial π/2-pulse prepares a coherent superposition of spin-up and spin-down states, enabling interference.

By varying the relative orientation of the spin-flip axes, different closed paths on the Bloch sphere are generated, resulting in different geometric phases. Since both spin components experience identical dynamical conditions in an ideal setup, their dynamical phases cancel in the relative phase measured by the polarimeter. The remaining phase-shift is therefore purely geometric.

The accumulated geometric phase is detected by scanning an additional phase-shifting coil, leading to sinusoidal intensity oscillations at the detector. A horizontal shift of these oscillations corresponds directly to the geometric phase. Measurements are performed for different rotation schemes of the spin flippers (single, parallel, and opposite rotations), allowing the dependence of the geometric phase on the relative rotation angle to be systematically investigated and compared to theoretical predictions.

3 Experimental setup

The experiment is performed at a polarized neutron beamline and is based on a neutron polarimetry setup designed to manipulate and analyze the spin state of neutrons. A schematic overview of the setup is shown in Figure 2.

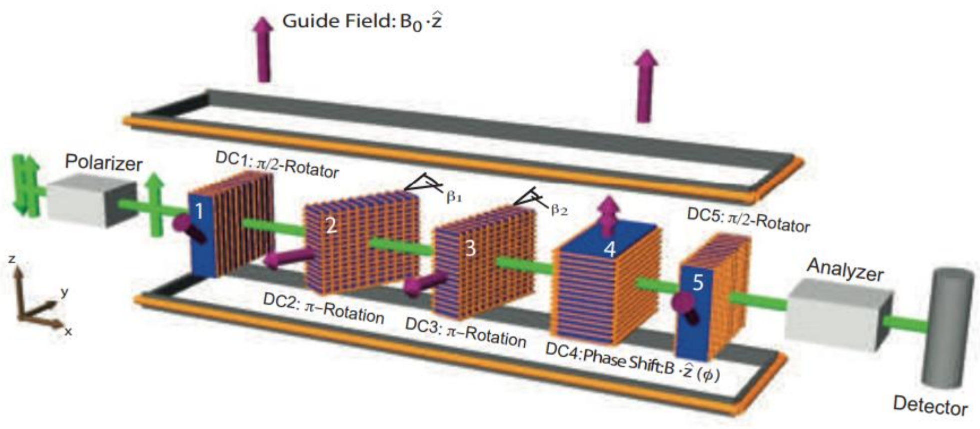

Figure 2 Schematic layout of the neutron polarimetry setup showing the polarizer, DC spin-rotator coils, phase-shifting coil, analyzer, and detector.

A monochromatic neutron beam is first polarized using a magnetic supermirror polarizer, which transmits neutrons with one spin orientation while suppressing the opposite spin state. The polarized neutron beam is then guided through the apparatus by a weak homogeneous magnetic guide field, which preserves the spin orientation and defines the quantization axis.

The manipulation of the neutron spin is achieved using a sequence of DC magnetic coils acting as spin rotators. These coils generate controlled magnetic fields that induce well-defined spin rotations via Larmor precession. The setup includes coils producing π/2- and π-rotations about selectable axes, allowing for precise control of the spin state and the realization of closed evolution paths on the Bloch sphere. By adjusting the relative orientation of the spin-flip axes, different geometric configurations of the spin evolution can be implemented.

An additional DC coil is used as a phase-shifting element. By scanning the current through this coil, a controlled relative phase between the spin components is introduced, enabling the detection of interference fringes in the neutron intensity.

After the spin manipulation, the neutron beam passes through a second supermirror, which acts as a spin analyzer and projects the final spin state onto a fixed measurement basis. The transmitted neutrons are detected by a neutron detector, where the intensity is recorded as a function of the applied phase. The measured intensity oscillations form the basis for the determination of the geometric phase.

4 Experimental procedure

4.1 Coil adjustment

4.1.1 Adjustment of DC2 and DC3 (π-flips)

Purpose

The purpose of this step is to adjust the DC2 and DC3 coils such that they perform accurate π-spin-flips. Proper adjustment is required to achieve a maximum inversion of the neutron spin and a high flipping ratio, which is essential for reliable polarimetric measurements.

Procedure

- Scans of the neutron-intensity as a function of the magnetic field component B_x generated by DC2 and DC3 are performed.
- Pronounced intensity minima in the B_x-scans, which indicate efficient spin inversion, are identified.
- Scans of the neutron intensity as a function of the magnetic field component B_z are performed for the previously identified B_x-scan-minima.
- The B_z-scan-minima is identified and fixed.
- The B_x-scan are repeated after the final adjustment to confirm the optimized configuration.

Plots and results

	DC2	DC3
I (B_x) [mA]	-1130	1261
I (B_z) [mA]	498	387
flipping ratio	18.63	15.74

Table 1 Identified currents and filliping ratio of DC2 and DC3 for π-rotation

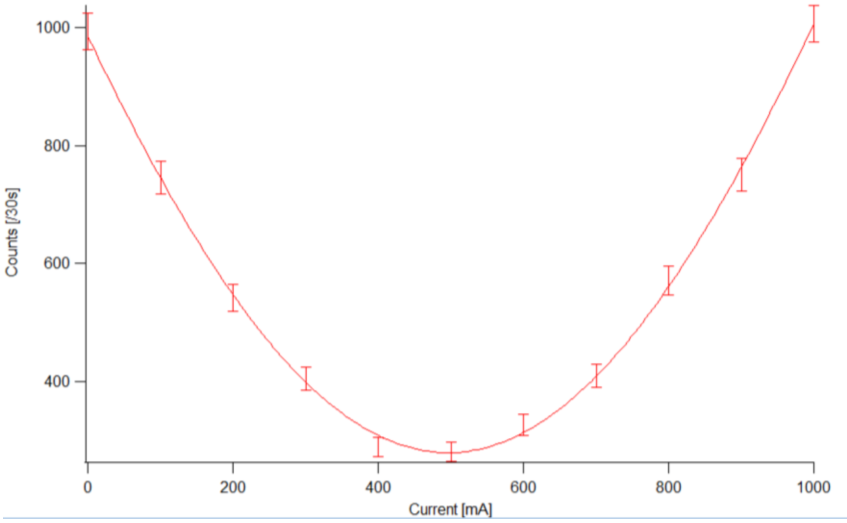

Figure 3 B_z Intensity Scan (DC2)

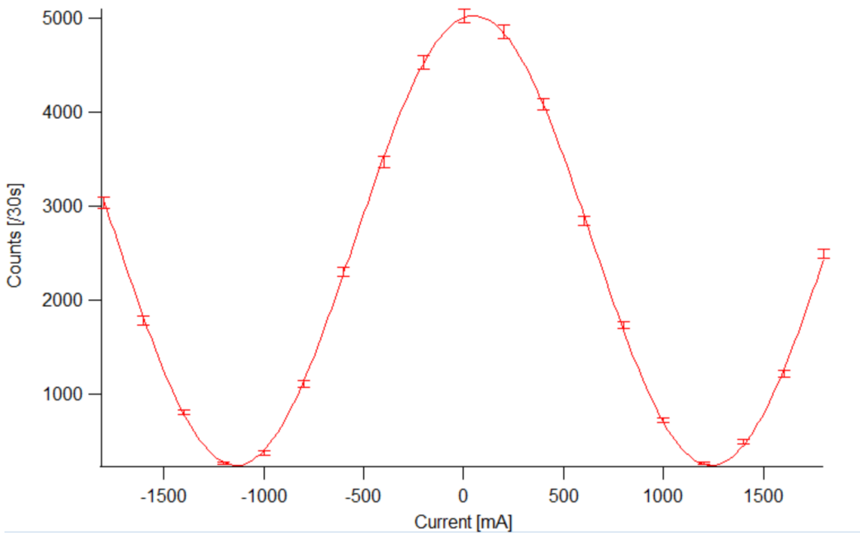

Figure 4 Final B_x Intensity Scan (DC2)

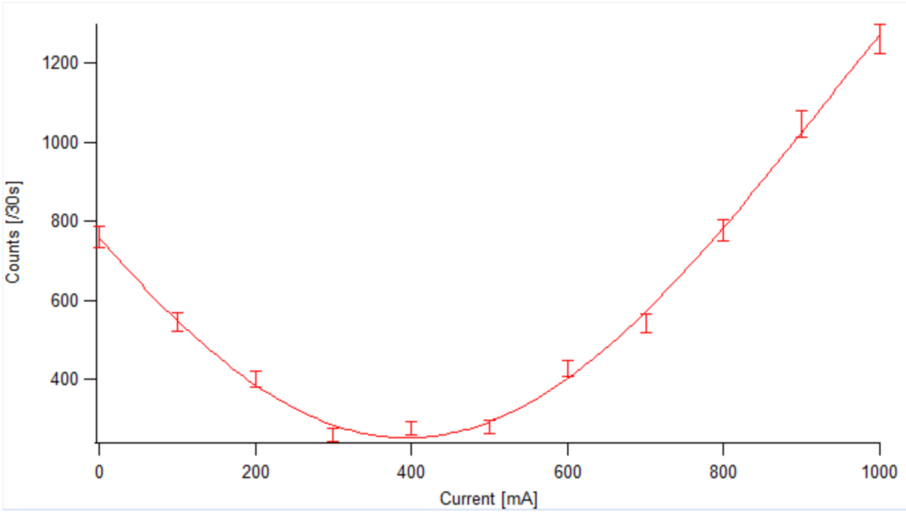

Figure 5 B_z Intensity Scan (DC3)

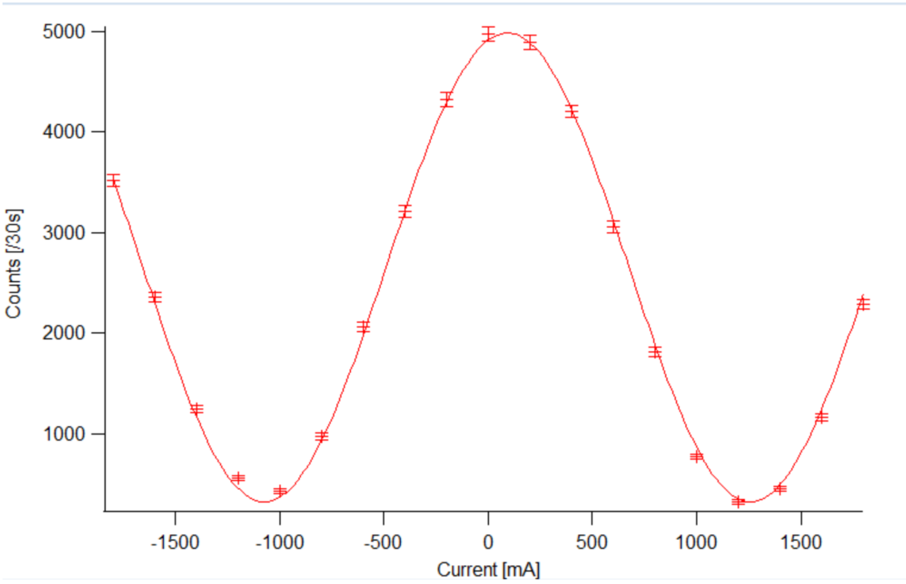

Figure 6 Final B_x Intensity Scan (DC3)

4.1.2 Adjustment of DC1 and DC5 (π/2-rotations)

Purpose

The purpose of this step is to determine the coil currents for DC1 and DC5 that correspond to π/2-spin-rotations. These rotations are required to prepare and recombine coherent superposition states of the neutron spin, enabling interference effects in the polarimeter.

Procedure

- Scans of the neutron intensity are performed as a function of the magnetic field component B_x generated by DC1.
- The oscillatory behavior of the neutron intensity resulting from spin precession is observed.
- The current value at which the intensity lies approximately midway between a maximum and a minimum (corresponding to a π/2-rotation of the spin) is identified.
- The current of DC1 is fixed at this value.
- The same procedure is repeated for DC5.

Plots and results

	DC1	DC5
I (B_x) [mA]	705	703

Table 2 Identified currents of DC1 and DC5 for π/2-rotation

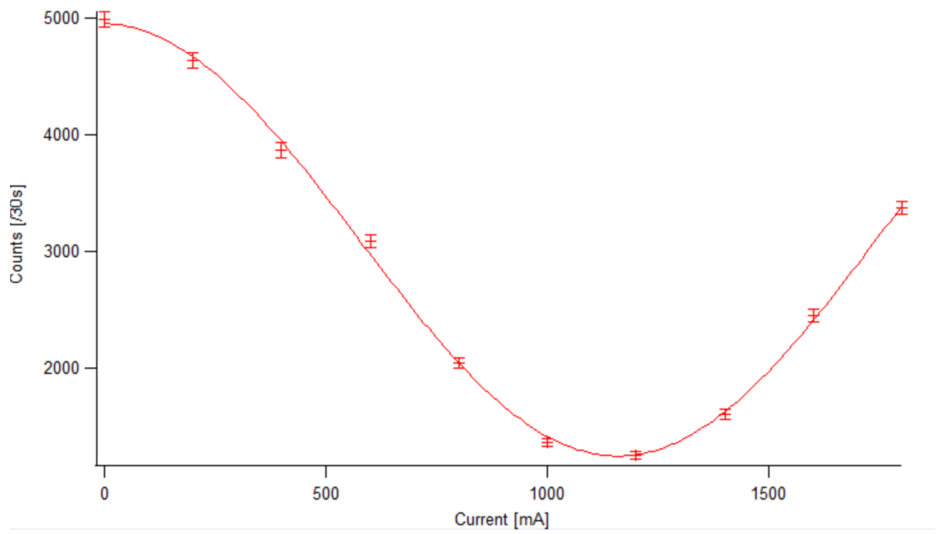

Figure 7 B_x Intensity Scan (DC1)

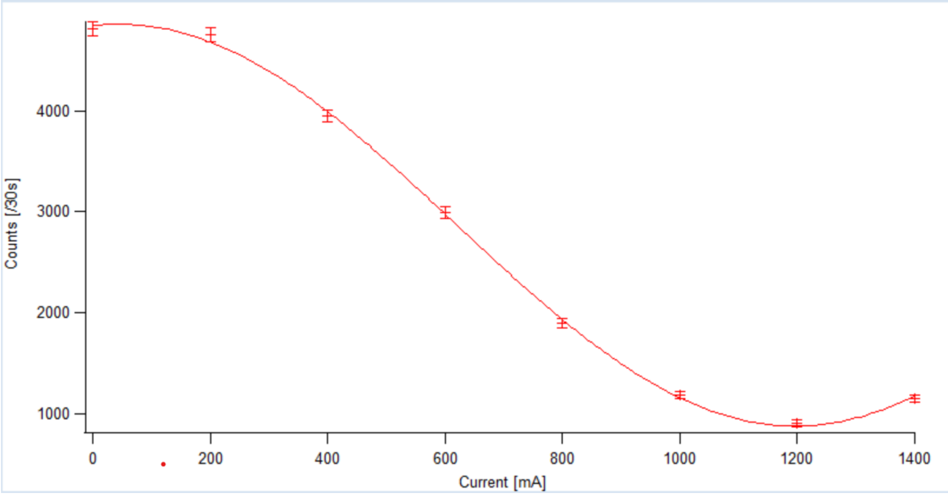

Figure 8 B_x Intensity Scan (DC5)

4.1.3 Adjustment of DC4 (phase-shifting coil)

Purpose

DC4 is used to generate a magnetic field in the z-direction in order to induce a controllable relative phase between the orthogonal spin eigenstates. The purpose of this step is to determine suitable scan parameters such that one full interference period is sampled by 11 measurement points.

Procedure

- A preliminary scan of the neutron intensity as a function of the DC4 current is performed.
- The periodic modulation of the neutron intensity is observed and the current difference corresponding to one full oscillation period.
- The scan increment is determined by dividing the period length by 11.

Result

The final DC4 increment scan parameter was determined to be 180 mA/step. This value was used for all following experiments.

5 Data analysis

5.1 Sinusoidal fits of the DC4 intensity oscillations

Purpose

The purpose of this step is the extraction of the phase parameter from each recorded DC4 intensity oscillation. The fitted phase represents the experimentally accessible quantity from which the geometric phase is determined. For the analysis, the program Igor Pro was used.

Procedure

- The recorded data sets of neutron intensity as a function of the DC4 current are imported into the data analysis program.
- Statistical uncertainties are assigned to all data points based on counting statistics, using the square root of the detected counts. These uncertainties are included as error bars in all plots.
- For each DC4 scan, a sinusoidal fit of the form is performed.
- The phase parameter ϕ for each fit is automatically calculated by the program.

Additional remark:

In addition to the standard measurement sequence, a second single rotation data set was recorded with the rotation angle β_2 kept fixed while varying only β_1. This additional measurement was performed to reduce potential systematic uncertainties arising from repeated manual adjustment or miscalibration of the β_2-rotator and to assess the influence of such effects on the extracted phase shift.

Plots and results

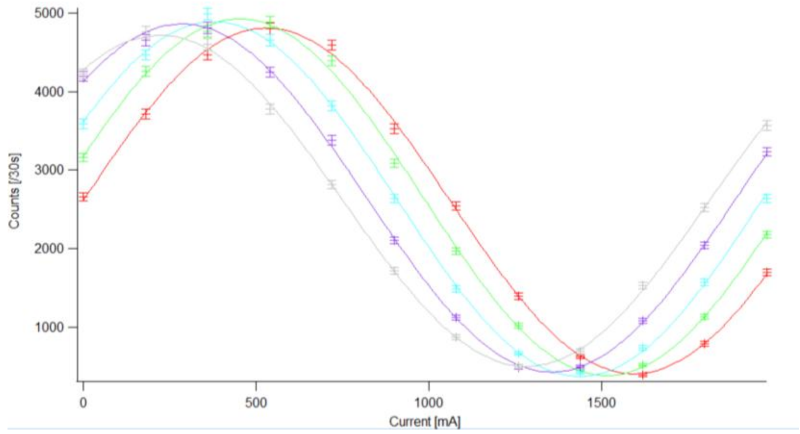

Figure 9 Intensity oscillation for single rotation (for fixed β_1)

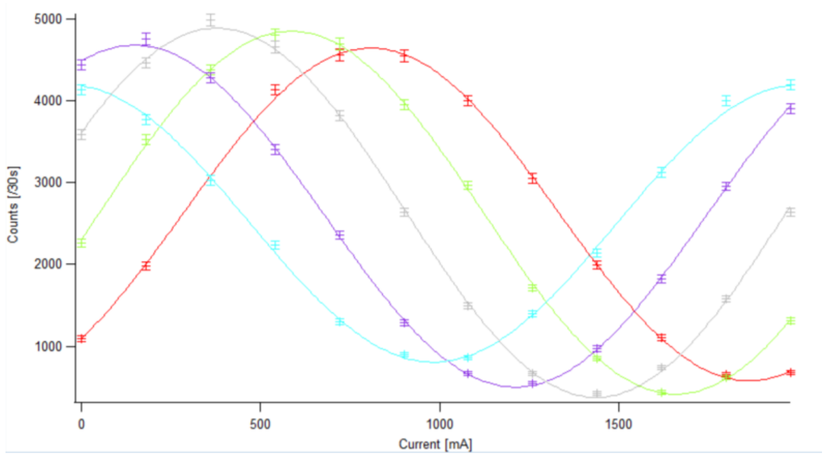

Figure 10 Intensity oscillation for opposite rotation

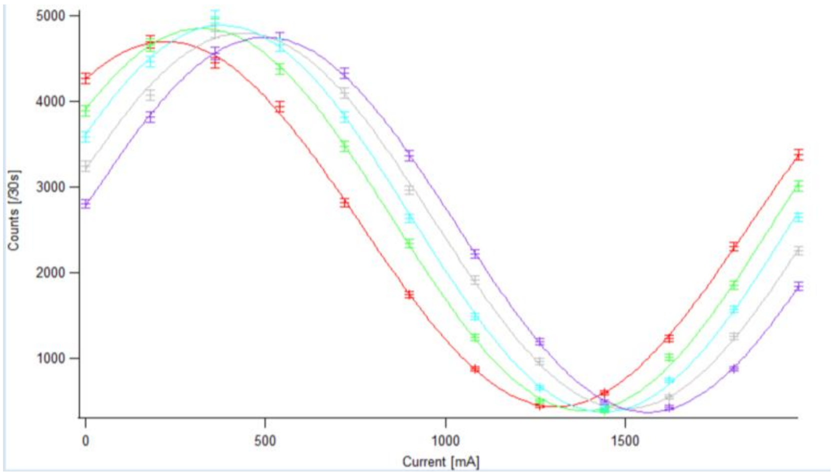

Figure 11 Intensity oscillation for parallel rotation

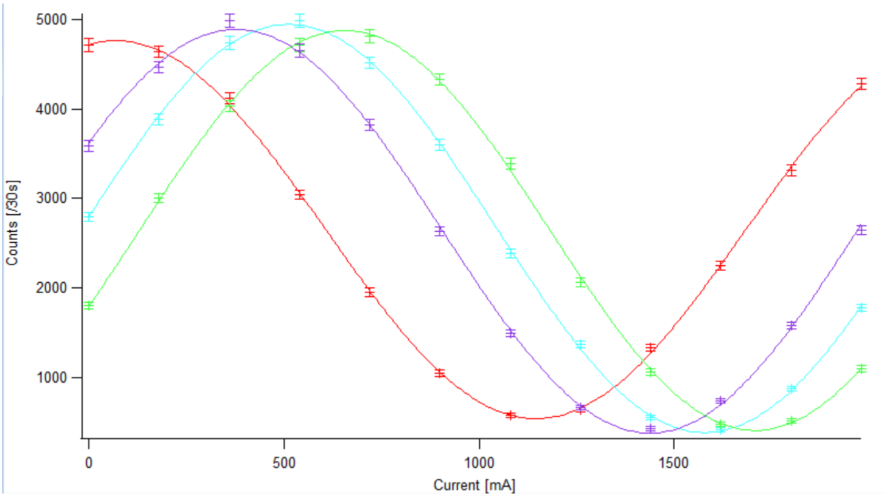

Figure 12 Intensity oscillation for single rotation (for fixed β_2)

Please note:

The dataset for the single rotation of $\beta_1 = 10°$ was somehow lost during the transfer from the computer to the USB-stick. The phase value, however, was documented during the experiment and is therefor included in *Table 6.*

5.2 Phase shift as a function of the relative angle $\delta\beta$

Purpose

The purpose of this step is the investigation of the dependence of the measured phase shift on the relative orientation of the spin-flip axes of DC2 and DC3 for different rotation schemes.

Procedure

- For each measurement, the relative angle
$$\delta\beta = \beta_1 - \beta_2$$
between the rotation axes of DC2 and DC3 is determined.
- Each extracted phase value ϕ is assigned to the corresponding value of $\delta\beta$.
- Separate data sets are compiled for the three rotation schemes: single, parallel, and opposite rotation.

Plots and results

β_1	β_2	calculated phase value and error ($\phi \pm \Delta\phi$) [rad]
0°	-20°	0.91221±0.0245
0°	-10°	0.73177±0.0234
0°	0°	0.45067±0.0209
0°	10°	0.23137±0.025
0°	20°	0.00381±0.021

Table 3 calculated phase value and error for single rotation (for fixed β_1)

β_1	β_2	calculated phase value and error ($\phi \pm \Delta\phi$) [rad]
-20°	20°	-0.83826±0.0224
-10°	10°	-0.14708±0.014
0°	0°	0.45067±0.0209
10°	-10°	1.1332±0.0228
20°	-20°	1.6915±0.0666

Table 4 calculated phase value and error for opposite rotation

β_1	β_2	calculated phase value and error ($\phi \pm \Delta\phi$) [rad]
-20°	-20°	0.097478±0.0837
-10°	-10°	0.27512±0.0217
0°	0°	0.45067±0.0209
10°	10°	0.61395±0.0181
20°	20°	0.92135±0.0212

Table 5 calculated phase value and error for parallel rotation

β_1	β_2	calculated phase value and error $(\phi \pm \Delta\phi)$ [rad]
-20°	0°	-0.39054±0.013
-10°	0°	0.055486±0.122
0°	0°	0.45067±0.0209
10°	0°	0.86205±0.0174
20°	0°	1.367±0.026

Table 6 calculated phase value and error for single rotation (for fixed β_2)

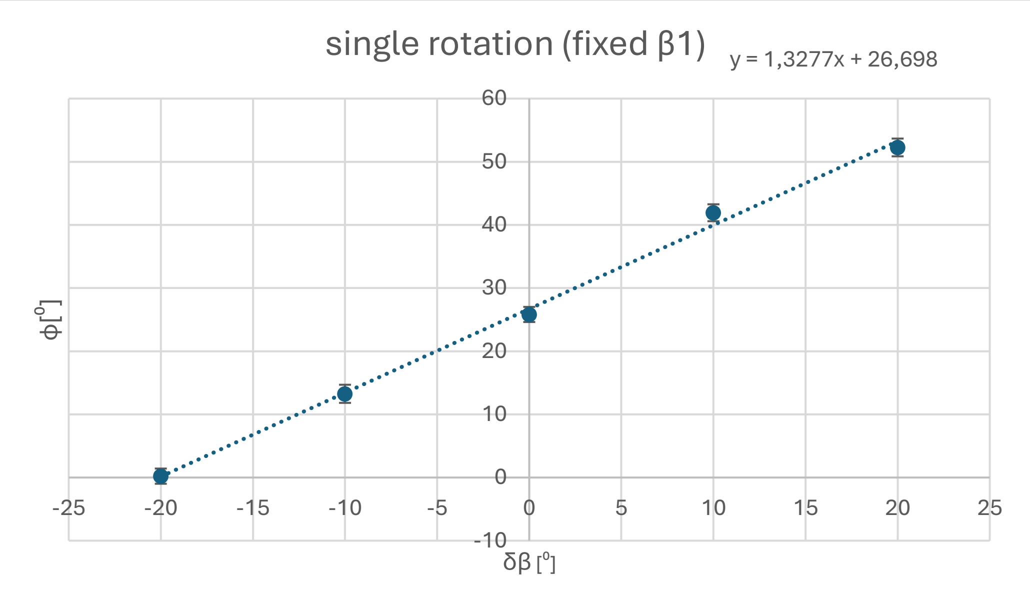

Figure 13 Phase shift relative to δβ for single rotation (for fixed β_1) including error and slope

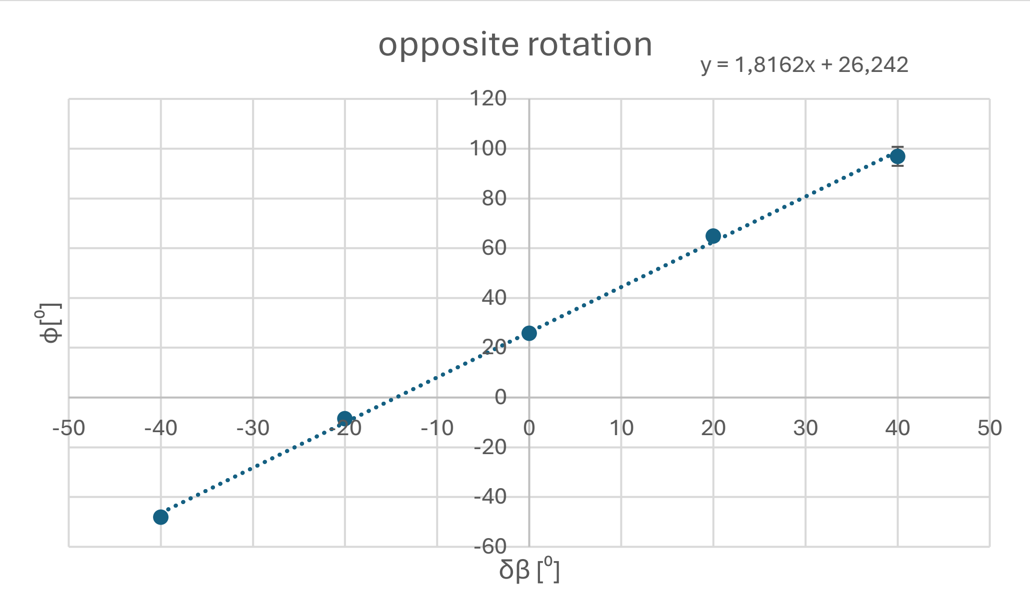

Figure 14 Phase shift relative to δβ for opposite rotation including error and slope

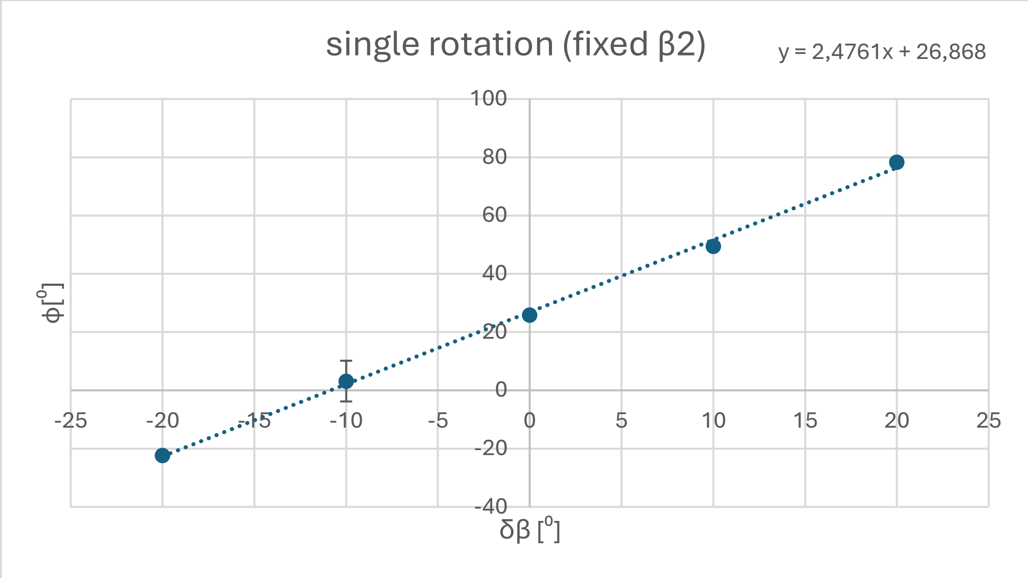

Figure 15 Phase shift relative to δβ for single rotation (for fixed β_2) including error and slope

<u>Please note:</u>

For the plots and the calculation of the slopes, the phase was converted from radian to degrees. For the case of parallel rotation, no such graph can be created, since $\delta\beta = \beta_1 - \beta_2$ always comes out to be zero because the angles are always kept identically to one another.

5.3 Determination of slopes and comparison with theory

Purpose

The purpose of this step is the quantitative determination of the geometric phase dependence on the relative rotation angle and its comparison with the theoretical prediction for a spin-½ system.

Procedure

- For each of the three phase-versus-angle data sets, Microsoft Excel created a linear fit of the form
$$\phi(\delta\beta) = m * \delta\beta + b.$$
- The slopes m can be extracted from the graphs in *5.2*.
- The experimentally obtained slopes are compared to the theoretical expectation for the geometric phase in a polarimetric measurement.

Plots and results

	calculated slope m	theoretical value	deviation [%]
single (fixed β_1)	1.3277	2	33.62
opposite	1.8162	2	9.19
single (fixed β_2)	2.4761	2	23.81

Table 7 calculated slope of single and opposite rotation

The theoretical value for the Berry phase in a polarimetric measurement of a spin-½ system is given by
$$\gamma = 2\,\delta\beta,$$
since the relative phase between the two spin components is measured. Consequently, a linear dependence of the measured phase shift ϕ on the relative rotation angle $\delta\beta$ with a slope of 2 is expected whenever $\delta\beta$ is varied. For the parallel rotation scheme, $\delta\beta = 0$ for all measurements and no geometric phase is expected.

6 Interpretation

The experimentally determined slopes for the different rotation schemes deviate by up to approximately 34% from the theoretical value of 2 expected for the geometric phase of a spin-½ system. In addition to typical experimental uncertainties, such as limited angular resolution of the rotators, reproducibility of the manually adjusted angles β_1 and β_2, and general alignment tolerances, these effects alone are insufficient to explain the observed systematic deviations.

The dominant source of the discrepancy is attributed to the fact that the neutron beam does not intersect the rotation axes of the spin-flip coils perfectly. A slight lateral displacement of the beam with respect to the coil axes results in unequal effective path lengths between the spin rotators. As a consequence, an additional dynamical phase is accumulated during the free precession of the neutron spin, which is not completely canceled by the polarimetric sequence. This residual dynamical contribution adds to or subtracts from the geometric phase, depending on the specific rotation scheme and orientation of the coils.

Mathematical description of the error caused by a lateral displacement of the neutron beam with respect to the coil axes, resulting in different effective path lengths between the coils:

$$The\ dynamical\ phase\ of\ a\ neutron\ spin\ in\ a\ guide\ field\ B_0 \parallel z$$
$$is\ given\ by\ the\ Larmor\ precession\ frequency$$

$$\omega_L = \frac{2|\mu|B_0}{\hbar}.$$

$$For\ a\ flight\ path\ L\ and\ neutron\ velocity\ v, the\ accumulated\ phase\ reads$$

$$\phi(L) = \omega_L t = \omega_L \frac{L}{v}.$$

$$If\ the\ path\ lengths\ between\ the\ spin\ rotators\ are\ not\ equal,$$

$$L_{12} \neq L_{23},$$

$$an\ additional\ dynamical\ phase\ arises:$$

$$\Delta\phi_{\text{dyn}} = \frac{2|\mu|B_0}{\hbar v}\,(L_{12} - L_{23}).$$

This interpretation is supported by the two independent single-rotation measurements. In these measurements, the extracted slopes are shifted in opposite directions relative to the theoretical value: one single rotation yields a slope reduced by approximately 0.7, while the second single rotation exhibits an increase of about 0.5. These opposite shifts are consistent with an additional dynamical phase contribution whose sign depends on the relative geometry of the neutron trajectory and the rotation axes. When applying the opposite rotation scheme, these contributions partially cancel, resulting in a significantly smaller deviation of about 0.2 from the theoretical slope, in good qualitative agreement with the observed result.

7 Conclusion & Summary

This experiment used neutron polarimetry in a Ramsey-type interferometric setup to study the geometric phase of a spin-½ neutron. The measurements show a clear linear dependence of the accumulated phase on the relative rotation angle, confirming the geometric origin of the effect. Deviations from the ideal theoretical prediction highlight the high sensitivity of geometric phase measurements to experimental alignment and systematic effects. Overall, the experiment offers important insights into the practical limitations of precision neutron polarimetry as well as the physical concept of geometric phases.

8 Appendix

Figures

Tables